

Note also that

$$\begin{aligned}
 \ell^* &\leq \frac{2}{4s+d} \log_2 \left(\frac{m+n}{\ln(\ln(m+n))} \right) + 1 \\
 &\leq \frac{2}{4s+d} \log_2(m+n) + 1 \\
 &\leq \left(\frac{2}{d \ln(2)} + 1 \right) \ln(m+n) \\
 &< 4 \ln(m+n)
 \end{aligned}$$

as $\ln(\ln(m+n)) > 1$ and $\ln(m+n) > 1$. We get

$$\begin{aligned}
 \ln \left(\frac{1}{w_{\lambda^*}} \right) &= 2 \ln(\ell^*) + \ln \left(\frac{\pi^2}{6} \right) \\
 &\leq 2 \ln(4 \ln(m+n)) + \ln \left(\frac{\pi^2}{6} \right) \\
 &\leq \left(2 \ln(4) + 1 + \ln \left(\frac{\pi^2}{6} \right) \right) \ln(\ln(m+n)) \\
 &< 5 \ln(\ln(m+n))
 \end{aligned}$$

as $\ln(\ln(m+n)) > 1$. Combining those upper bounds, we get

$$\begin{aligned}
 \frac{\ln(\frac{1}{\alpha}) + \ln(\frac{1}{w_{\lambda^*}})}{(m+n) \sqrt{\lambda_1^* \dots \lambda_d^*}} &\leq \frac{1}{(m+n) \sqrt{\lambda_1^* \dots \lambda_d^*}} \left(\ln \left(\frac{1}{\alpha} \right) + 5 \ln(\ln(m+n)) \right) \\
 &\leq \left(\ln \left(\frac{1}{\alpha} \right) + 5 \right) \frac{\ln(\ln(m+n))}{m+n} \frac{1}{\sqrt{\lambda_1^* \dots \lambda_d^*}} \\
 &\leq 2^{\frac{d}{2}} \left(\ln \left(\frac{1}{\alpha} \right) + 5 \right) \frac{\ln(\ln(m+n))}{m+n} \left(\frac{m+n}{\ln(\ln(m+n))} \right)^{d/(4s+d)} \\
 &= 2^{\frac{d}{2}} \left(\ln \left(\frac{1}{\alpha} \right) + 5 \right) \left(\frac{m+n}{\ln(\ln(m+n))} \right)^{-4s/(4s+d)}
 \end{aligned}$$

as $\ln(\ln(m+n)) > 1$. Note also that

$$\ell^* \geq \frac{2}{4s+d} \log_2 \left(\frac{m+n}{\ln(\ln(m+n))} \right)$$

giving

$$(\lambda_i^*)^{2s} = (2^{-\ell^*})^{2s} \leq \left(\frac{m+n}{\ln(\ln(m+n))} \right)^{-4s/(4s+d)}$$

for $i = 1, \dots, d$. Hence, we get

$$\sum_{i=1}^d (\lambda_i^*)^{2s} \leq d \left(\frac{m+n}{\ln(\ln(m+n))} \right)^{-4s/(4s+d)}.$$

We obtain

$$\begin{aligned} \rho(\Delta_\alpha^{\Lambda^w, B_{1:3}}, \mathcal{S}_d^s(R), \beta, M)^2 &\leq C_6(M, d, s, R, \beta) \left(\sum_{i=1}^d (\lambda_i^*)^{2s} + \frac{\ln(\frac{1}{\alpha}) + \ln(\frac{1}{w_{\lambda^*}})}{(m+n) \sqrt{\lambda_1^* \cdots \lambda_d^*}} \right) \\ &\leq C_7(M, d, s, R, \alpha, \beta)^2 \left(\frac{m+n}{\ln(\ln(m+n))} \right)^{-4s/(4s+d)} \end{aligned}$$

where $C_7(M, d, s, R, \alpha, \beta) := \sqrt{C_6(M, d, s, R, \beta) \max \left\{ d, 2^{\frac{d}{2}} \left(\ln(\frac{1}{\alpha}) + 5 \right) \right\}}$. We conclude that

$$\rho(\Delta_\alpha^{\Lambda^w, B_{1:3}}, \mathcal{S}_d^s(R), \beta, M) \leq C_7(M, d, s, R, \alpha, \beta) \left(\frac{m+n}{\ln(\ln(m+n))} \right)^{-2s/(4s+d)}.$$

Hence, the test $\Delta_\alpha^{\Lambda^w, B_{1:3}}$ is optimal in the minimax sense up to an iterated logarithmic term. Since it does not depend on the unknown parameters s and R , our aggregated MMDAgg test $\Delta_\alpha^{\Lambda^w, B_{1:3}}$ is minimax adaptive over the Sobolev balls $\{\mathcal{S}_d^s(R) : s > 0, R > 0\}$.

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